

UTILITIES



GateOverflow Chatbot — Demo Kit

Source: DEMO_SCRIPT.md

Everything you need to present the app: a **ready-made narrated video**, an **editable wireframe**, a **storyboard/narration script**, and a set of **sample Q&A** to type live.

■ What's in **demo/**

File	What it is	How to use
<code>GateOverflow-Chatbot-Demo.mp4</code>	~4 min narrated demo, 18 scenes (1080p, voiced) from real app footage — every feature + differentiation + GateOverflow PYQ links + site awareness + AI Coach + adaptive practice + save/export	Play it directly in the meeting / embed in slides
<code>GateOverflow-Chatbot-Feature-Guide.mp4</code>	~9 min detailed guide (28 scenes) from real screenshots of the running app — incl. AI Coach , adaptive weak-area practice , rank band / percentile trend , plan-adherence , GateOverflow PYQ links, platform awareness, bookmarks & history, cheat-sheet & PDF export, sketch, multilingual, Socratic, dislike→regenerate — per feature: purpose, example, benefit vs ChatGPT/Claude, how to use	Train the team / onboarding deep-dive
<code>live/</code>	Real screenshots captured from the running app (source stills for the guide video)	Reuse in slides/docs
<code>build_live_video.py</code>	Rebuilds the detailed guide from <code>live/</code> screenshots (run the app first, recapture, then this)	Regenerate the real-footage video
<code>GateOverflow-Chatbot-Demo.docx</code>	Full documentation : features, workflow, architecture, differentiation, sample Q&A, roadmap	Hand to the team / present alongside the video
<code>Why-GateOverflow-Chatbot.docx</code>	Shorter positioning doc (why it beats general chatbots)	Quick read / leave-behind
<code>wireframe.html</code>	Annotated wireframe + UX spec (screens, flow, components)	Open in a browser; Print → Save as PDF; or import to Figma
<code>build_video.py</code> / <code>build_demo_doc.py</code>	Regenerate the video / docx from scratch (offline)	<code>python demo/build_video.py</code> · edit <code>SCENES</code> / sections to customise
<code>frames/</code>	Per-scene PNGs + clips (build artifacts)	Reuse stills in slides

Regenerate the video anytime: `python demo/build_video.py` (needs Google Chrome + ffmpeg + macOS `say` — all already on this machine). Swap the voice via the `VOICE` constant, or edit any scene's `narration` / `body`.

■ A. Live-demo storyboard (≈ 5 min)

Run the real app first:

BASH

```
# terminal 1 – backend (needs OPENAI_API_KEY in .env, index built)
uvicorn api.main:app --host 0.0.0.0 --port 8000 --reload
# terminal 2 – frontend
cd frontend && npm install && npm run dev      # → http://localhost:5173
```

#	Scene	Do this	Say this (narration)
1	Intro	Show the welcome screen	“This is the GateOverflow Chatbot — an AI tutor for GATE CS and DA, grounded in real previous-year questions.”
2	Greeting	Type 	“It handles small talk naturally — a greeting gets a friendly welcome, not a wrong answer.”
3	Concept Q&A	Ask “ <i>Explain conflict serializability with an example.</i> ”	“Ask anything in the syllabus. Answers stream live with proper maths, and cite the source material.”
4	Citations	Expand the source card	“Every answer shows where it came from — file, page and a match score.”
5	Follow-up	Click a follow-up chip (e.g. <i>Give me a similar PYQ</i>)	“Follow-up chips keep the conversation going with one click.”
6	PYQ practice	Ask “ <i>Give me a GATE CS PYQ on TLB effective access time and solve it.</i> ”	“It has ingested thousands of PYQs — it poses the question and walks through the solution.”
7	Quiz	Click  Quiz me	“It can run an interactive quiz, one question at a time, and score you.”
8	Attachment	 attach a screenshot of a question (or paste an image)	“Stuck on a question in an image or PDF? Attach it — the vision model reads it and solves it.”
9	Feedback	 a reply, pick a reason, add a correction	“Rate any answer. A dislike opens a quick form and feeds the learning loop.”
10	Regenerate (RLHF)	Click ↻ regenerate	“Regenerate re-answers, steered by your feedback — and stores a preference pair for offline RLHF training.”
11	Export	Open http://localhost:8000/export/preferences	“All feedback exports as JSONL, ready for reward-model or DPO fine-tuning.”
12	Theme	Toggle dark/light	“Polished, responsive UI in light and dark — built to production

#	Scene	Do this	Say this (narration)
			standards.”

■ B. Sample questions to type (copy-paste)

These showcase breadth + the grounded/citation behaviour:

1. **Hi** — (*greeting handling*)
2. **Explain conflict serializability with a precedence-graph example.**
3. **Solve: a paged system has TLB hit ratio 0.9, TLB access 10 ns, memory access 100 ns. Find the effective access time.**
4. **Derive the time complexity of merge sort using the Master Theorem.**
5. **What is the bias-variance trade-off and how does regularisation help?** (*LaTeX-heavy*)
6. **Give me a GATE DA previous-year question on probability, then explain it step by step.**
7. **Quiz me with 5 mixed GATE CS MCQs, one at a time, and score me at the end.**
8. **What can you do?** — (*capabilities*)
9. **Tell me a joke about cricket.** — (*out-of-scope → polite decline*)

Example interaction (what the bot returns)

You: Solve: TLB hit ratio 0.9, TLB access 10 ns, memory access 100 ns — effective access time?

GO Buddy:

- **TLB hit** (prob 0.9): $10 + 100 = \mathbf{110\ ns}$
- **TLB miss** (prob 0.1): $10 + 100 + 100 = \mathbf{210\ ns}$
- **EAT** = $0.9 \cdot 110 + 0.1 \cdot 210 = \mathbf{120\ ns}$ ✓

Sources: filter1_volume2.pdf · p. 57 · GATA-Data · 88% match

■ C. Feature checklist (mention these)

- ✓ **RAG** over your PDFs / notes (CS · DS · ML · AI) with citations
- ✓ **Previous-year questions** ingested (GATE CS/DA, ISRO, NIELIT, UGC-NET, TIFR)
- ✓ **Streaming** answers · **LaTeX** maths · syntax-highlighted code
- ✓ **Per-answer actions**: copy · like · dislike · regenerate
- ✓ **Follow-up suggestions** + **interactive quiz**
- ✓ **Attachments**: image (vision) + PDF/text (extracted to context)

-  **Greeting / out-of-scope** handling
 -  **Feedback** → **RLHF**: preference pairs + JSONL export for DPO/reward-model
 -  **Persistence**: SQLite (Postgres-ready) — conversations, messages, feedback
 -  **Stack**: React + TS + Vite + Tailwind ⇔ FastAPI ⇔ Chroma + OpenAI
 -  **Production-ready**: Docker, tests, configurable via `.env`, GateOverflow branding
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■ D. Turn the wireframe into a Figma file

1. Open `wireframe.html` in Chrome.
2. In Figma, install the “**html.to.design**” plugin.
3. Plugin → *Import* → paste the local file (or a hosted URL) → it imports editable layers matching these screens.

(Alternatively: Print the page to PDF and drop it into any slide deck.)